

HAT-P-19b

R-2033

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_110616 · created 2026-08-08T01:32:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455728.9326 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.022
Transit depth (Rp/Rs)^2	3.25 +/- 0.81 [%]
Orbital Inclination (inc)	86.78 +/- 0.81
Airmass coefficient 1 (a1)	1.42 +/- 0.026
Airmass coefficient 2 (a2)	-0.1185 +/- 0.0057
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	None
Optimal Aperture	2.96
Optimal Annulus	8.43
Transit Duration (day)	0.087 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.022 vs 0.1418 (deviation 1.74σ)
Duration fit vs published	0.087 d vs 0.1196 d (deviation 1.55σ)
Mid-time O–C	1.3 min
Depth SNR	8.2
Residual scatter	1.82 %

Light curve

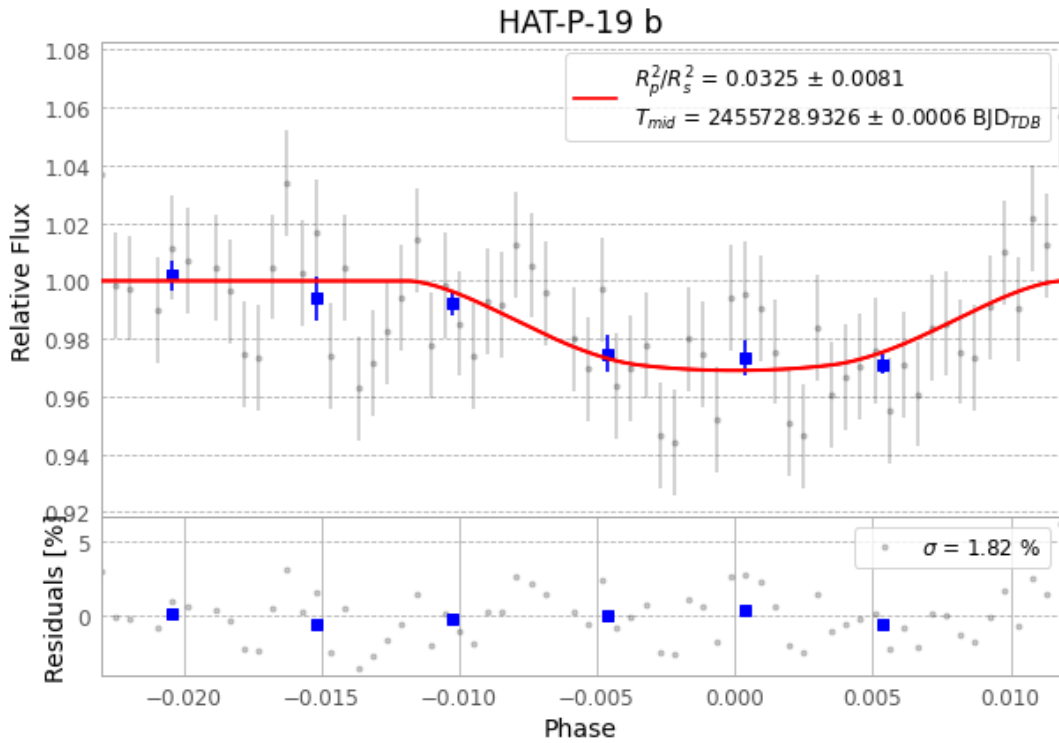


Figure 1 — FinalLightCurve_HAT-P-19 b_2011-06-16.png (SHA-256 6120c285240e02e3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 207], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 51f11bfda323e092...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (50 MB), HATP-19110616081213.FITS → HATP-19110616120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-110616.log (3986099fa255...), FinalLightCurve_HAT-P-19 b_2011-06-16.png (6120c285240e...), FinalLightCurve_HAT-P-19 b_2011-06-16.csv (dd00aa3c3325...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)